XLV National Mathematics Olympiad: 3rd round, April 1997

Problem 1. Find all natural numbers a, b and c such that the roots of the equation

$$\begin{aligned} x^2 - 2ax + b &= 0 \\ x^2 - 2bx + c &= 0 \\ x^2 - 2cx + a &= 0 \end{aligned}$$

are natural numbers.

Problem 2. Given a convex quadrilateral $ABCD$ which can be inscribed in a circle. Let F be the intersecting point of diagonals AC and BD and E be the intersecting point of the lines AD and BC. If M and N are the midpoints of AB and CD, prove that

$$\frac{MN}{EF} = \frac{1}{2} \cdot \left| \frac{AB}{CD} - \frac{CD}{AB} \right|.$$

Problem 3. Prove that the equation

$$x^2 + y^2 + z^2 + 3(x + y + z) + 5 = 0$$

has no solution in rational numbers.

Problem 4. Find all continuous functions $f(x)$ defined in the set of real numbers and such that

$$f(x) = f\left(x^2 + \frac{1}{4}\right)$$

for all real x.

Problem 5. Two squares K_1 and K_2 with centres M and N and sides of length 1 are placed in the plane in a way that $MN = 4$, two of the sides of K_1 are parallel to MN and one of the diagonals of K_2 lies on the line MN. Find the locus of midpoints of segments XY where X is an interior point for K_1 and Y is an interior point for K_2.

Problem 6. Find the number of non-empty sets of $S_n = \{1, 2, \ldots, n\}$ such that there are no two consecutive numbers in one and the same set.

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